Financial Fraud Detection

Exploratory Data Analysis (EDA)

Task 1: Find the dimensions (number of rows and columns) of the dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
data = pd.read_csv("cc_data.csv")  # Replace "your_dataset.csv" with the path to your dataset
# dropping index column
data=data.drop(columns='index')
# Dimensions of the dataset
print("Dimensions of the dataset:", data.shape)
```

Task 2: Check if unique values are present in each categorical variable

```
# Unique values in each categorical variable
unique_values = data.select_dtypes(include=['object']).nunique()
print("Unique values in each categorical variable:")
print(unique_values)
```

Task 3: Find the distribution of numerical variables in the dataset

```
# Distribution of numerical variables
numerical_summary = data.describe()
print("Distribution of numerical variables:")
numerical_summary
```

Task 4: Check if there is any missing values in the dataset

```
# Missing values in the dataset
missing_values = data.isnull().sum()
print("Missing values in the dataset:")
print(missing_values)
```

Task 5: Find the summary statistics (mean, median, min, and max) for numerical variables

```python
# Summary statistics for numerical variables
summary_statistics = data.describe(include='all')
print("Summary statistics for numerical variables:")
summary_statistics
```

Task 6: Check if there is any correlation between numerical variables. If so, how strong is the correlation

```python
# Correlation between numerical variables
correlation_matrix = data.select_dtypes([float,int]).corr()
print("Correlation between numerical variables:")
correlation_matrix
```

Task 7: Check if the distribution of an amt differ across is_fraud categories

```python
# How does the distribution of a amt differ across is_fraud categories?
sns.boxplot(x='is_fraud', y='amt', data=data)
plt.show()
```

Task 8: Check if there is any outliers in the city_pop and amt

```python
# Outliers in numerical variables
sns.boxplot(data=data[['amt']])
plt.show()

sns.boxplot(data=data[['city_pop']])
plt.show()
```

Task 9: Check for trends or patterns in the data over time (if applicable)

```python
# Trends or patterns in the data over time
# This depends on the specific data you're working with. You can plot time-series data and look for trends.

plt.figure(figsize=(14,6))
plt.plot(data['trans_date_trans_time'],data['amt'])
plt.show()
```